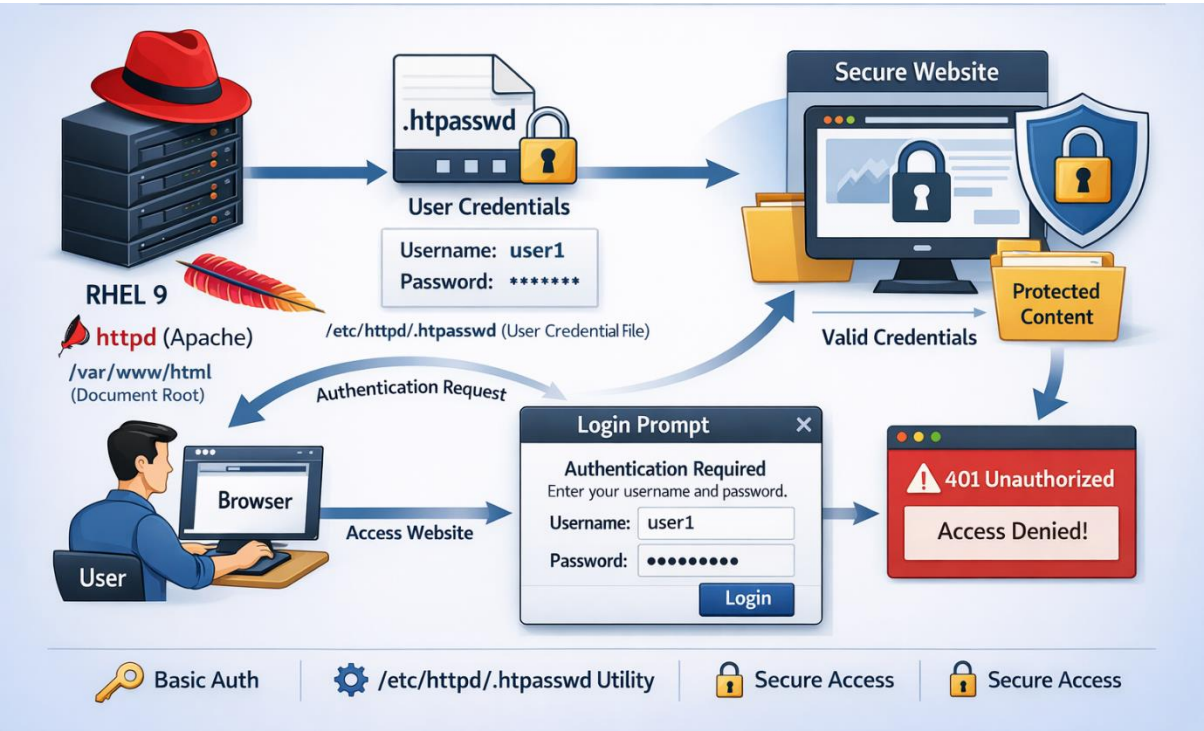
Secure Apache Web Server with Basic Authentication on RHEL 9



This project focuses on configuring and securing an Apache web server on RHEL 9 by implementing Basic HTTP Authentication. The primary objective is to restrict unauthorized access to web content by requiring users to authenticate with valid credentials before accessing protected resources.

The project workflow encompasses:

1. Apache Web Server Setup

Installing and configuring the Apache HTTP Server (httpd) on a RHEL 9 system. The service is enabled and started using systemctl, ensuring the web server is accessible and operational.

2. Web Content Configuration

Creating and managing web content under the default document root directory (/var/www/html). A sample webpage is deployed to verify the successful setup of the Apache server.

3. Authentication Setup using htpasswd

Implementing Basic HTTP Authentication using the htpasswd utility. A password file is created to securely store user credentials, allowing only authorized users to access the protected content.

4. Apache Configuration for Security

Modifying Apache configuration files to enable authentication. This includes defining protected directories and applying authentication directives such as:

- AuthType Basic
- AuthName
- AuthUserFile
- Require valid-user

5. Access Control and Testing

Testing the configuration by accessing the web server through a browser. Users are prompted to enter a username and password, ensuring that unauthorized users are denied access.

❖ Project Objective

In real-world environments, web servers often host sensitive or restricted content that must be protected from unauthorized users. Without proper security mechanisms, anyone can access the hosted data.

Basic HTTP Authentication provides a simple yet effective way to secure web resources by enforcing user verification. This project demonstrates how to implement authentication at the web server level, ensuring controlled access and improved security for hosted applications.

❖ Key Learning Outcomes

- Apache (httpd) installation and configuration on RHEL 9
- Implementing Basic Authentication using htpasswd
- Securing web directories with Apache configuration
- Understanding access control mechanisms
- Practical experience in Linux-based web server security

❖ Prerequisites

- RHEL 9 system installed and running
- Root or sudo user access
- Active internet connection

Implementation Steps

Step 1: Install Apache Web Server (httpd)

Installed the Apache HTTP Server (httpd) package using the dnf package manager along with its required dependencies to set up the web server environment.

```
[root@rhel ~]#
[root@rhel ~]# dnf install httpd
Updating Subscription Management repositories.
Red Hat Enterprise Linux 9 for x86_64 - AppStream (RPMs)                                                    4.0 MB/s |  85 MB     00:21
Red Hat Enterprise Linux 9 for x86_64 - BaseOS (RPMs)                                                        12 MB/s | 107 MB     00:09
Last metadata expiration check: 0:00:18 ago on Thursday 19 March 2026 10:24:53 PM.
Dependencies resolved.
================================================================================================================================================
 Package                                 Architecture            Version                              Repository                                Size
================================================================================================================================================
Installing:
 httpd                                   x86_64                  2.4.62-7.el9_7.3                     rhel-9-for-x86_64-appstream-rpms           47 k
Installing dependencies:
 apr                                     x86_64                  1.7.0-12.el9_3                       rhel-9-for-x86_64-appstream-rpms          126 k
 apr-util                                x86_64                  1.6.1-23.el9                         rhel-9-for-x86_64-appstream-rpms           97 k
 apr-util-bdb                            x86_64                  1.6.1-23.el9                         rhel-9-for-x86_64-appstream-rpms           14 k
 httpd-core                              x86_64                  2.4.62-7.el9_7.3                     rhel-9-for-x86_64-appstream-rpms          1.5 M
 httpd-tools                             x86_64                  2.4.62-7.el9_7.3                     rhel-9-for-x86_64-appstream-rpms           81 k
 redhat-logos-httpd                      noarch                  90.5-1.el9_6.1                       rhel-9-for-x86_64-appstream-rpms           16 k
Installing weak dependencies:
 apr-util-openssl                        x86_64                  1.6.1-23.el9                         rhel-9-for-x86_64-appstream-rpms           17 k
 mod_http2                               x86_64                  2.0.26-5.el9                         rhel-9-for-x86_64-appstream-rpms          163 k
 mod_lua                                 x86_64                  2.4.62-7.el9_7.3                     rhel-9-for-x86_64-appstream-rpms           59 k

Transaction Summary
================================================================================================================================================
Install  10 Packages

Total size: 2.1 M
Total download size: 270 k
Installed size: 6.0 M
```

Step 2: Start and Enable Apache Service

Started the Apache service and enabled it to run automatically at system boot. Verified the service status to ensure the web server is active and running properly.

```
[root@rhel ~]# systemctl start httpd
[root@rhel ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@rhel ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Thu 2026-03-19 22:26:12 IST; 11s ago
       Docs: man:httpd.service(8)
   Main PID: 35385 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 16394)
     Memory: 41.8M (peak: 42.1M)
        CPU: 212ms
     CGroup: /system.slice/httpd.service
             ├─35385 /usr/sbin/httpd -DFOREGROUND
             ├─35386 /usr/sbin/httpd -DFOREGROUND
             ├─35387 /usr/sbin/httpd -DFOREGROUND
             ├─35388 /usr/sbin/httpd -DFOREGROUND
             └─35409 /usr/sbin/httpd -DFOREGROUND

Mar 19 22:26:12 rhel systemd[1]: Starting The Apache HTTP Server...
Mar 19 22:26:12 rhel httpd[35385]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 10.78.238.89. Set the 'ServerName' dire
Mar 19 22:26:12 rhel httpd[35385]: [Thu Mar 19 22:26:12.488763 2026] [core:error] [pid 35385:tid 35385] (EAI 2)Name or service not known: AH00549: Failed to resolve se
Mar 19 22:26:12 rhel httpd[35385]: [Thu Mar 19 22:26:12.514148 2026] [core:error] [pid 35385:tid 35385] (EAI 2)Name or service not known: AH00549: Failed to resolve se
Mar 19 22:26:12 rhel httpd[35385]: Server configured, listening on: port 80
Mar 19 22:26:12 rhel systemd[1]: Started The Apache HTTP Server.
lines 1-24/24 (END)
```

Step 3: Configure Firewall to Allow HTTP Traffic

Configured the firewall to allow HTTP (port 80) traffic using firewalld and reloaded the firewall to apply the changes.

```
[root@rhel ~]# firewall-cmd --add-service=http --permanent
success
[root@rhel ~]# firewall-cmd --reload
success
[root@rhel ~]#
```

Why: Port 80 is required because Apache serves web pages over HTTP, and the firewall blocks incoming traffic by default, so enabling it allows users to access the website.

Step 4: Verify Firewall Configuration

Verified that the HTTP service is successfully added to the firewall rules using firewall-cmd --list-all.

```
[root@rhel ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client http ssh
  ports: 30/tcp 80/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@rhel ~]#
```

Step 5: Create and Verify Web Page Content

Created a custom HTML file (index.html) inside /var/www/html/ to host web content and verified its content using the cat command.

```
[root@rhel ~]# cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
    <title>Secure Apache Server</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #eef2f7;
            text-align: center;
            padding-top: 80px;
        }
        .box {
            background: #ffffff;
            padding: 30px;
            margin: auto;
            width: 50%;
            border-radius: 10px;
            box-shadow: 0px 0px 10px rgba(0,0,0,0.2);
        }
        h1 {
            color: #2c3e50;
        }
        .highlight {
            color: #2980b9;
            font-weight: bold;
        }
        p {
            color: #555;
            font-size: 18px;
        }
        .footer {
            margin-top: 20px;
            padding: 10px;
            background-color: #2980b9;
            color: white;
            border-radius: 5px;
            display: inline-block;
        }
    </style>
</head>
<body>

<div class="box">
    <h1>Secure Apache Web Server</h1>
```

Step 6: Install httpd-tools Package

Installed the httpd-tools package using dnf, which provides the htpasswd utility required for creating and managing user authentication credentials. This step is necessary because Apache does not include this tool by default, and it is essential for implementing Basic Authentication.

```
[root@rhel ~]# dnf install httpd-tools
Updating Subscription Management repositories.
Last metadata expiration check: 0:16:44 ago on Thursday 19 March 2026 10:24:53 PM.
Dependencies resolved.
================================================================================================================================================================
 Package                                Architecture                Version                                  Repository                                      Size
================================================================================================================================================================
Installing:
 httpd-tools                            x86_64                      2.4.62-7.el9_7.3                         rhel-9-for-x86_64-appstream-rpms                 81 k
Installing dependencies:
 apr                                    x86_64                      1.7.0-12.el9_3                           rhel-9-for-x86_64-appstream-rpms                126 k
 apr-util                               x86_64                      1.6.1-23.el9                             rhel-9-for-x86_64-appstream-rpms                 97 k
 apr-util-bdb                           x86_64                      1.6.1-23.el9                             rhel-9-for-x86_64-appstream-rpms                 14 k
Installing weak dependencies:
 apr-util-openssl                       x86_64                      1.6.1-23.el9                             rhel-9-for-x86_64-appstream-rpms                 17 k

Transaction Summary
================================================================================================================================================================
Install  5 Packages

Total download size: 335 k
Installed size: 736 k
Is this ok [y/N]: y
Downloading Packages:
(1/5): apr-util-bdb-1.6.1-23.el9.x86_64.rpm                                                                                   11 kB/s |  14 kB     00:01
(2/5): apr-util-openssl-1.6.1-23.el9.x86_64.rpm                                                                               13 kB/s |  17 kB     00:01
(3/5): apr-util-1.6.1-23.el9.x86_64.rpm                                                                                       74 kB/s |  97 kB     00:01
(4/5): httpd-tools-2.4.62-7.el9_7.3.x86_64.rpm                                                                               155 kB/s |  81 kB     00:00
(5/5): apr-1.7.0-12.el9_3.x86_64.rpm                                                                                         226 kB/s | 126 kB     00:00
----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                        183 kB/s | 335 kB     00:01
Running transaction check
Transaction check succeeded.
```

Step 7: Create User Credentials File

Created a secure password file /etc/httpd/.htpasswd using the htpasswd command and added a user (admin). This file stores encrypted user credentials and is used by Apache to verify users before granting access to protected resources.

```
[root@rhel ~]# htpasswd -c /etc/httpd/.htpasswd admin
New password:
Re-type new password:
Adding password for user admin
[root@rhel ~]#
```

Step 8: Configure Apache for Basic Authentication

Configured Apache by creating an authentication file (auth.conf) to protect the /var/www/html directory. Enabled Basic Authentication and linked it with the .htpasswd file so that only authenticated users can access the web content.

```
[root@rhel ~]# cat /etc/httpd/conf.d/auth.conf
<Directory "/var/www/html">
    AuthType Basic
    AuthName "Restricted Area"
    AuthUserFile /etc/httpd/.htpasswd
    Require valid-user
</Directory>
[root@rhel ~]#
```

Configuration Explanation :

- **<Directory "/var/www/html">**

  → Tells Apache which folder needs to be protected

- **AuthType Basic**

  → Enables username and password login system

- **AuthName "Restricted Area"**

  → Shows a message in the login popup

- **AuthUserFile /etc/httpd/.htpasswd**

  → Points to the file where usernames and passwords are stored

- **Require valid-user**

  → Allows only registered users to access the website

Step 9: Restart Apache Service

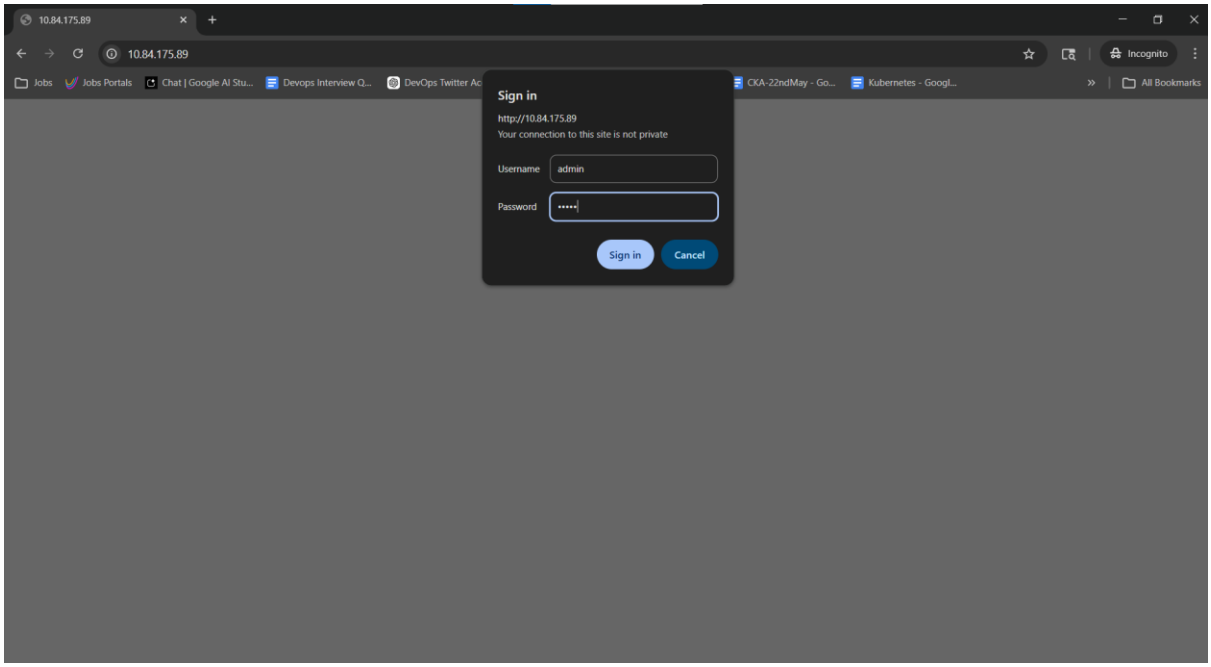
Restarted the Apache service to apply the new authentication configuration. This ensures that all changes made in the configuration files are loaded and the server starts enforcing authentication rules.

```
[root@rhel ~]# systemctl restart httpd
[root@rhel ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Thu 2026-03-19 23:25:38 IST; 8s ago
       Docs: man:httpd.service(8)
   Main PID: 39173 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 16394)
     Memory: 31.9M (peak: 32.2M)
        CPU: 203ms
     CGroup: /system.slice/httpd.service
             ├─39173 /usr/sbin/httpd -DFOREGROUND
             ├─39174 /usr/sbin/httpd -DFOREGROUND
             ├─39175 /usr/sbin/httpd -DFOREGROUND
             ├─39176 /usr/sbin/httpd -DFOREGROUND
             └─39177 /usr/sbin/httpd -DFOREGROUND

Mar 19 23:25:37 rhel systemd[1]: Starting The Apache HTTP Server...
Mar 19 23:25:38 rhel httpd[39173]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::a00:27ff:fec1:d792%enp0s3. Set t
Mar 19 23:25:38 rhel httpd[39173]: [Thu Mar 19 23:25:38.086089 2026] [core:error] [pid 39173:tid 39173] (EAI 2)Name or service not known: AH00549: Failed to resolve s
Mar 19 23:25:38 rhel httpd[39173]: [Thu Mar 19 23:25:38.111994 2026] [core:error] [pid 39173:tid 39173] (EAI 2)Name or service not known: AH00549: Failed to resolve s
Mar 19 23:25:38 rhel httpd[39173]: Server configured, listening on: port 80
Mar 19 23:25:38 rhel systemd[1]: Started The Apache HTTP Server.
lines 1-24/24 (END)
```

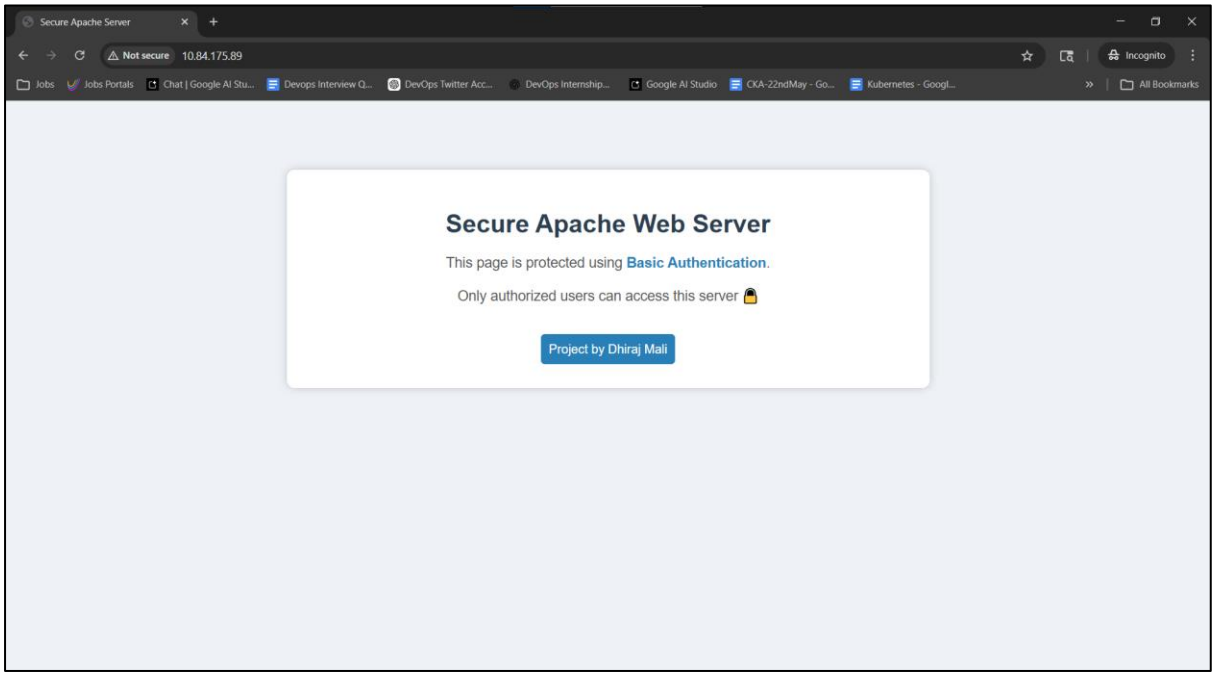
Step 10: Test Authentication in Browser

Accessed the web server through a browser, which prompted for a username and password. This confirms that the authentication mechanism is active and the server is restricting access as configured.



Step 11: Verify Protected Web Page Access

Entered valid credentials to successfully access the secured web page. This verifies that only authorized users can view the content, confirming that Basic Authentication is working correctly.



❖ Conclusion

The project successfully demonstrated the setup and security of an Apache Web Server on RHEL 9 using Basic Authentication. It ensured that only authorized users could access the web content and provided practical knowledge of web server configuration and access control.